



LUTHFIE BUDIE ANDIKA

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Portfolio: luthfiebudie.github.io/portfolio-test/

A Computer Science graduate with a strong motivation to develop professional skills and experience. Experience designing autonomous prototypes, integrating computer vision, and implementing machine learning and pathfinding algorithms on hardware. Possesses strong analytical and creative problem-solving skills to master unfamiliar technologies quickly, combined with clear technical communication skills to present complex ideas effectively.

PROJECTS & WORK EXPERIENCE

Oct 2023 – Jan 2024	KOMINFO - Data Analyst & IT Support Intern Developed and optimized robotic systems using machine learning (Perceptron, MLP, Backpropagation) to improve performance, while analyzing data using Microsoft Excel to support data-driven decision-making.
May 2025 – Jun 2025	Thesis Project - A Star Line Follower Car Repository: https://github.com/LuthfieBudie/A-Star-Algorithm-Smart-Car - Algorithm Integration: Implemented A* pathfinding algorithm for real-time autonomous routing, optimizing robot navigation efficiency by 15%. - Robotic Development: Designing the hardware architecture, integrating the line sensor circuit with motor actuators, and configuring the car's movement control to maintain precision. - Embedded Systems: Developing control programming logic using C++ language in the Arduino IDE environment for efficient processing of electronic component input-output data.
Dec 2025 – Feb 2026	Independent Project – Smart Anti Sleep Helmet Repository: https://github.com/LuthfieBudie/Smart-Anti-Sleep-Helmet - Computer Vision Integrated sensor cameras for real-time face/eye tracking and feature extraction to recognize signs of drowsiness with high accuracy. - AI Development: Implementing Machine Learning models (Perceptron / Backpropagation) to classify driver status and trigger safety alert systems. - Embedded Systems: Connecting camera and sensor modules to helmet hardware to process lightweight computing data in a portable manner.
May 2026 – May 2026	Independent Project - Real time AI thrift store simulation pipeline Repository: https://github.com/LuthfieBudie/Real-time-AI-thrift-store-simulation-pipeline - AI Development: Building a Machine Learning-based simulation pipeline using the Backpropagation algorithm to perform data analysis and automate inventory classification, improving data processing efficiency by 20%. - Data Analytics: Processing over 1,000 rows of transaction simulation data, optimizing database queries, and creating visualizations of pipeline system performance metrics.

TECHNICAL SKILLS, TOOLS & LANGUAGE

Programming & Databases: Python, C++, SQL, Arduino IDE, HTML, CSS
Data & Intelligent System: Data Analyst, Machine Learning, Computer Vision, Tableau, PowerBI, Kaggle
Design & System Modeling: Figma, UI/UX Design, Ms Visio, Draw.io
Creative & Productivity Tools: Adobe Photoshop, Illustrator, Premiere Pro, After Effects, Filmora, Microsoft Office (Word, Excel, PowerPoint)
Languages: Bahasa Indonesia (Native), English (Intermediate)

EDUCATION

Jul 2021 – Jul 2025	Computer Science – State Islamic University of North Sumatra Graduated cum laude (GPA 3.53) and has experience in various technical projects, including robotics, systems, and website development, including collaborative team-based projects and a final project presented at an international workshop, as well as participation in competency development seminars and webinars.
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CERTIFICATIONS

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| • Junior Graphic Designer | • Canva Graphic Design Integration |
| • UI/UX Design Certificate | • Intro to Data Analytics |